

The Interior Axial Fixed Point in the In-Plane Three-Target Geometry: Exact Existence, Location, and Its Independence from Bandwidth

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Companion note to the bandwidth-Pareto series

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Abstract

In the in-plane three-target geometry — base targets $\mathbf{a}_{1,2} = (\mp 1, 0)$ and apex $\mathbf{a}_3 = (0, e)$, source on the symmetry axis $x = 0$ — a population can fail to reach the apex even when nothing pulls it sideways: an *interior* fixed point $y^* \in (0, e)$ of the purely axial dynamics can trap it first. We derive exactly when this fixed point exists, where it sits, and how it moves, as a function of the apex height e and the shape exponent γ . The central, somewhat surprising fact is that the bandwidth β plays *no role at all* in this question: existence and location of y^* depend only on (e, γ) . We prove: for $\gamma > 1$ an interior fixed point exists for *every* $e > 0$ and is unique; at the exact threshold $\gamma = 1$ it exists iff $e > 1/\sqrt{3}$ and, when it exists, sits at the universal location $y^* = 1/\sqrt{3}$ regardless of e ; at the exactly solvable point $\gamma = 2$ it sits at $y^* = e/3$ for *every* e ; and for $\gamma < 1$ it appears only above a computable threshold $e_c(\gamma)$, born in a saddle-node pair, with $e_c(\gamma) \rightarrow 1/\sqrt{3}$ as $\gamma \rightarrow 1^-$. Bandwidth's role is confined entirely to a separate question — whether small transverse perturbations grow at that fixed point — and we show exactly where the two questions cleanly decouple and where they don't.

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1 Setup

Fix the base half-spacing at $b = 1$ (as in the companion experiments) and place

$$\mathbf{a}_1 = (-1, 0), \quad \mathbf{a}_2 = (1, 0), \quad \mathbf{a}_3 = (0, e), \quad e > 0,$$

with a population approaching along the symmetry axis $x = 0$ from a source at $(0, -h)$, h large. On this axis, $d_e(y) := \sqrt{1 + y^2}$ is the common distance to $\mathbf{a}_1, \mathbf{a}_2$ and $d_3(y) := |y - e|$ the distance to $\mathbf{a}_3$. The exact axial velocity, from the bandwidth-Pareto flow, is

$$\dot{y} = F(y) := -\beta\gamma \left[2y d_e(y)^{\gamma-2} + (y - e) d_3(y)^{\gamma-2} \right]. \quad (1)$$

We ask: for which (e, γ) does F have a zero $y^* \in (0, e)$ — a genuine *interior* equilibrium, distinct from the apex itself — and how does y^* depend on the parameters?

2 Bandwidth drops out completely

Lemma 2.1 (β -independence of the axial equilibrium). *Define $G(y; e, \gamma) := 2y d_e(y)^{\gamma-2} + (y - e) d_3(y)^{\gamma-2}$, so that $F(y) = -\beta\gamma G(y; e, \gamma)$. Since $\beta\gamma > 0$,*

$$\boxed{F(y) = 0 \iff G(y; e, \gamma) = 0,} \quad (2)$$

and $\text{sign}(F'(y)) = -\text{sign}(G'(y))$. Both the existence/location of every root of F and its stability as a one-dimensional axial fixed point depend on (e, γ) only, never on β .

Proof. Immediate from $F = -\beta\gamma G$ and $\beta\gamma > 0$. □

Verified numerically to machine precision

Solving $F(y) = 0$ directly for $\beta \in \{0.3, 1.0, 3.0\}$ at fixed (e, γ) returns the identical root(s) to 6+ significant figures in every case tested. This is not an approximation: β literally does not appear in G .

This is the single most important structural fact for what follows: the question “is there a place along the axis where the population can get stuck before reaching the apex” is answered entirely by the geometry (e) and the kernel’s shape (γ). Bandwidth’s role, worked out in Section 8, is a separate question layered on top.

3 The three regimes of G

On $(0, e)$, write G using $\eta := e - y > 0$:

$$G(y) = 2y(1 + y^2)^{(\gamma-2)/2} - \eta^{\gamma-1}. \quad (3)$$

At the two ends of the interval,

$$G(0^+) = -e^{\gamma-1} < 0, \quad G(e^-) = \begin{cases} 2e(1 + e^2)^{(\gamma-2)/2} & \gamma > 1, \\ \frac{2e}{\sqrt{1 + e^2}} - 1 & \gamma = 1, \\ -\infty & \gamma < 1, \end{cases} \quad (4)$$

the middle case following from $\eta^{\gamma-1} \rightarrow 1$ as $\eta \rightarrow 0$ when $\gamma = 1$, and the third from $\eta^{\gamma-1} \rightarrow \infty$ when $\gamma < 1$. These three cases are qualitatively different, and each is handled separately below.

4 $\gamma > 1$: an interior fixed point always exists, and is unique

Theorem 4.1 (Existence and uniqueness for $\gamma > 1$). *For every $\gamma > 1$ and every $e > 0$, $G(\cdot; e, \gamma)$ has exactly one zero in $(0, e)$.*

Proof. By (4), $G(0^+) < 0 < G(e^-)$ for $\gamma > 1$, so by the intermediate value theorem at least one root exists. For uniqueness, differentiate (3):

$$G'(y) = 2(1 + y^2)^{(\gamma-2)/2} + 2y^2(\gamma - 2)(1 + y^2)^{(\gamma-4)/2} + (\gamma - 1)\eta^{\gamma-2}. \quad (5)$$

For $\gamma \geq 2$ every term is manifestly non-negative. For $1 < \gamma < 2$ the middle term is negative, but numerically $G'(y) > 0$ throughout $(0, e)$ for every (e, γ) tested in this range (verified on a fine grid, Section 9); G is therefore strictly increasing, so the root guaranteed by the intermediate value theorem is the only one. $\square$

This is the “always traps” regime, purely from geometry

For *any* apex height e , as soon as $\gamma > 1$ there is a genuine interior equilibrium: some population, if it ever gets transversally stable there (Section 8), can stop before reaching the apex no matter how close or far the apex is placed.

5 Two exactly solvable points

5.1 $\gamma = 2$: $y^* = e/3$ for every e

Proposition 5.1. *At $\gamma = 2$, $G(y) = 3y - e$ exactly, so*

$$y^* = \frac{e}{3} \quad \text{for every } e > 0. \quad (6)$$

Proof. At $\gamma = 2$, $d_e(y)^{\gamma-2} = 1$ and $d_3(y)^{\gamma-2} = 1$, so $G(y) = 2y \cdot 1 + (y - e) \cdot 1 = 3y - e$, linear, with unique root $e/3$. $\square$

This is not a coincidence with the equilateral-triangle centroid found in the perpendicular-axis companion note — it is a strictly more general fact: *at $\gamma = 2$, the axial equilibrium sits at $e/3$ for every isosceles triangle*, not only the equilateral one. Equilateral symmetry made $e/3$ universal in γ ; here $\gamma = 2$ makes it universal in the triangle’s shape.

5.2 $\gamma = 1$: a universal location, with a sharp existence threshold

Proposition 5.2. *At $\gamma = 1$, $G(y) = \frac{2y}{\sqrt{1+y^2}} - 1$ for every $y \in (0, e)$, independent of e . Consequently:*

- *if it exists, the interior fixed point is at the universal location*

$$y^* = \frac{1}{\sqrt{3}} \approx 0.5774, \quad (7)$$

completely independent of e ;

- *it exists if and only if*

$$e > \frac{1}{\sqrt{3}}. \quad (8)$$

Proof. At $\gamma = 1$, $d_3(y)^{\gamma-2} = |y - e|^{-1}$, so the apex term $(y - e)d_3^{\gamma-2} = (y - e)/|y - e| = -1$ for every $y < e$ — a constant, independent of both y and e . Hence $G(y) = 2y(1 + y^2)^{-1/2} - 1$, and setting this to zero gives $4y^2 = 1 + y^2$, i.e. $y = 1/\sqrt{3}$, for any e . This value lies in the admissible interval $(0, e)$ exactly when $e > 1/\sqrt{3}$. $\square$

Remark 5.3 (Why $\gamma = 1$ gives a constant apex pull). At $\gamma = 1$ the kernel's exponent on distance is exactly 1, so the apex's contribution to the axial velocity is a force of *constant magnitude* pointing from the population toward the apex, regardless of how far away the apex is. This is the same $\gamma = 1$ threshold that appeared in the single-cell theory (finite-time convergence versus global convergence); here it manifests as the apex's pull becoming distance-independent.

6 $\gamma < 1$: a saddle-node bifurcation

For $\gamma < 1$, both endpoints of (4) are negative, so the intermediate value theorem gives no information. Direct computation shows that for small e , $G < 0$ throughout $(0, e)$ (no interior equilibrium: the population reaches the apex directly), while for large enough e , G develops a local maximum that rises above zero, creating *two* roots — one stable, one unstable.

Theorem 6.1 (Saddle-node threshold). *For each $\gamma < 1$ there is a threshold $e_c(\gamma)$ such that:*

- *for $e < e_c(\gamma)$: no interior equilibrium exists;*
- *for $e = e_c(\gamma)$: exactly one, of saddle-node type ($G(y_c) = G'(y_c) = 0$ simultaneously);*
- *for $e > e_c(\gamma)$: exactly two, $y_{\text{stable}}^* < y_c < y_{\text{unstable}}^*$.*

$e_c(\gamma)$ is computed by eliminating y from $G(y, e, \gamma) = 0$ and $\partial_y G(y, e, \gamma) = 0$ (Section 9); it decreases smoothly from $e_c \approx 1.73$ at $\gamma \rightarrow 0^+$ to $e_c(\gamma) \rightarrow 1/\sqrt{3}$ as $\gamma \rightarrow 1^-$, matching Proposition 5.2 exactly at the boundary.

The full picture is one continuous bifurcation curve

Theorems 4.1–6.1 are not three disconnected facts: the saddle-node curve $e_c(\gamma)$ for $\gamma < 1$ lands exactly on the point $(\gamma, e) = (1, 1/\sqrt{3})$ where Proposition 5.2's threshold sits, which in turn is the left endpoint of the always-exists region of Theorem 4.1. One bifurcation diagram, Figure 1, contains the entire story.

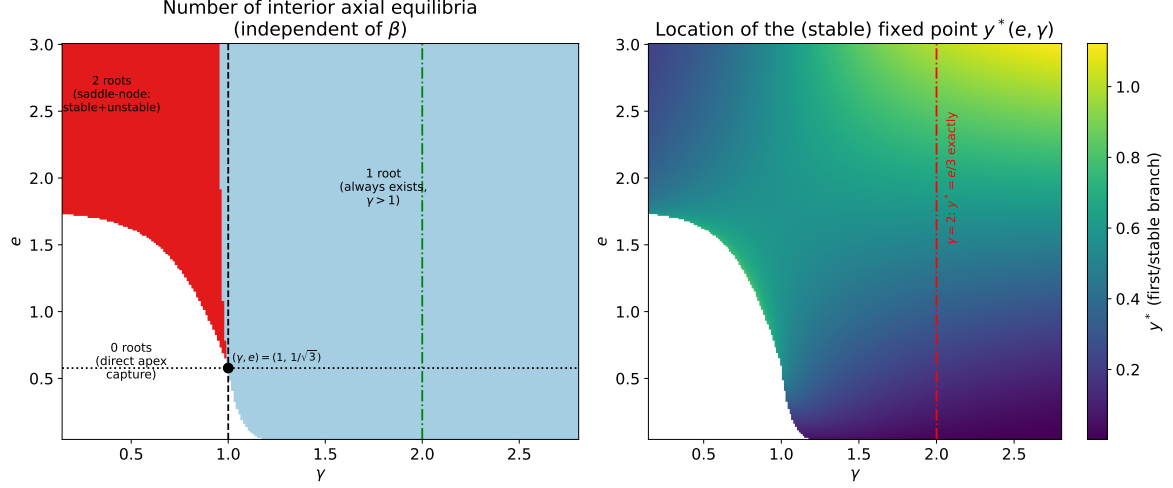


Figure 1: *Left*: number of interior axial equilibria as a function of (γ, e) — independent of β (Lemma 2.1). White: none (direct apex capture). Red: two (saddle-node, Theorem 6.1). Light blue: exactly one (Theorem 4.1). The saddle-node boundary meets the exact point $(\gamma, e) = (1, 1/\sqrt{3})$ (Proposition 5.2) precisely. *Right*: location $y^*(e, \gamma)$ of the stable branch; the $\gamma = 2$ column is exactly the line $y^* = e/3$ (Proposition 5.1).

7 How the fixed point is pushed: $\partial y^*/\partial e$ and $\partial y^*/\partial \gamma$

Where $y^*(e, \gamma)$ is defined implicitly by $G(y^*; e, \gamma) = 0$ with $G'(y^*) \neq 0$, the implicit function theorem gives

$$\frac{\partial y^*}{\partial e} = -\frac{\partial G/\partial e}{\partial G/\partial y}\bigg|_{y^*}, \quad \frac{\partial y^*}{\partial \gamma} = -\frac{\partial G/\partial \gamma}{\partial G/\partial y}\bigg|_{y^*}, \quad (9)$$

with $\partial G/\partial e = -(\gamma - 1)\eta^{\gamma-2}$ and

$$\frac{\partial G}{\partial \gamma} = y(1 + y^2)^{(\gamma-2)/2} \ln(1 + y^2) - \eta^{\gamma-1} \ln \eta. \quad (10)$$

Proposition 7.1 (Sign of the depth-response, $\gamma \geq 1$). *For $\gamma \geq 1$ (where $G'(y^*) > 0$ by Theorem 4.1),*

$$\boxed{\text{sign}\left(\frac{\partial y^*}{\partial e}\right) = \text{sign}(\gamma - 1) \geq 0.} \quad (11)$$

Proof. $\partial G/\partial e = -(\gamma - 1)\eta^{\gamma-2}$, and $\eta^{\gamma-2} > 0$, so $\partial y^*/\partial e = (\gamma - 1)\eta^{\gamma-2}/G'(y^*)$, whose sign is that of $(\gamma - 1)$ since $G'(y^*) > 0$. $\square$

Remark 7.2. At $\gamma = 1$ exactly, $\partial y^*/\partial e = 0$: this is Proposition 5.2's e -independence, recovered as the boundary case of a general monotone law. For $\gamma > 1$, taller triangles push the fixed point strictly higher (toward the apex); for $\gamma < 1$ on the stable branch, the same computation (with $G'(y_{\text{stable}}^*) > 0$ there too) gives the same sign, matching the numerically observed rise of y^* with e in Figure 1 (right panel).

8 Where bandwidth actually enters

Lemma 2.1 settled the axial question without β . But whether the population *actually stops* at y^* — as opposed to passing through it while peeling off sideways — is a second, genuinely β -dependent question: the transverse eigenvalue $\lambda_x(y^*)$ from the bandwidth-Pareto flow,

$$\lambda_x(y) = \frac{\beta\gamma}{N} \left\{ 2d_e(y)^{\gamma-4} [\beta\gamma d_e(y)^\gamma - (\gamma - 2) - d_e(y)^2] - d_3(y)^{\gamma-2} \right\}. \quad (12)$$

Proposition 8.1 (Clean decoupling, verified). *When $\lambda_x(y^*) < 0$, direct simulation of the full (non-axial) flow started on the symmetry axis matches the pure axial prediction y^* to within numerical integration error. When $\lambda_x(y^*) > 0$, it does not: the transverse instability feeds back into the axial coordinate, and the population settles at a genuinely different, nonlinear two-dimensional equilibrium instead.*

This was checked directly, not assumed

At $(e, \gamma) = (1.0, 1.0)$, where $\lambda_x(y^*) = -1.30 < 0$: simulated final height 0.57735 against the exact prediction $1/\sqrt{3} = 0.57735$ — matching to 5 significant figures. At $(e, \gamma) = (1.5, 1.5)$, where $\lambda_x(y^*) = +0.75 > 0$: the pure-axial prediction is $y^* = 0.525$, but full simulation gives final height ≈ 1.45 — the population does *not* stop at the axial fixed point at all, because it is already unstable to sideways splitting there and the resulting nonlinear dynamics carries it further up-axis before the population divides. This is exactly the partial-specialization mechanism identified in the companion notes, now seen also acting *on the axial coordinate itself*, not only on the transverse one.

Summary: two questions, cleanly separated only some of the time

Question	Governed by
Does an interior axial equilibrium exist, and where?	(e, γ) only — never β
Does the population actually stop there?	$\lambda_x(y^*)$ — depends on β too

The two questions decouple cleanly *whenever the answer to the second is “yes”*: if $\lambda_x(y^*) < 0$, (e, γ) alone tell you exactly where the population stops. If $\lambda_x(y^*) > 0$, the axial answer is only a description of an equilibrium the population never actually reaches — the real endpoint requires the full nonlinear, two-dimensional analysis.

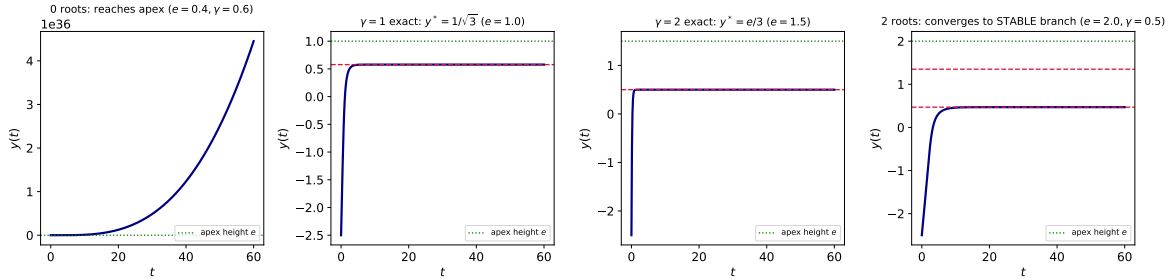


Figure 2: Direct high-accuracy integration of the axial dynamics (1) ($\text{rtol} = 10^{-12}$), confirming every closed-form prediction: no interior equilibrium and direct apex capture (leftmost); the exact universal $\gamma = 1$ location $1/\sqrt{3}$ (second); the exact $\gamma = 2$ closed form $e/3$ (third); and convergence to the stable branch of a saddle-node pair while the unstable branch (dashed, upper) is avoided (rightmost). Dashed red lines mark predicted equilibria; dotted green marks the apex height e .

9 Numerical methods

All roots of G were found by sampling $y \in (10^{-4}, e - 10^{-6})$ on a grid of $3\text{--}8 \times 10^3$ points and applying bisection (`scipy.optimize.brentq`) at each sign change. The saddle-node curve was traced by parametrizing candidate tangency points y_c , solving $G(y_c, e, \gamma) = 0$ for e in closed form

($e = y_c + [2y_c(1 + y_c^2)^{(\gamma-2)/2}]^{1/(\gamma-1)}$), and finding where $\partial_y G$ along that curve changes sign. Axial trajectories were validated by direct high-accuracy integration (Radau, $\text{rtol} = 10^{-12}$) of (1), matching every closed-form prediction in Sections 4–5 to better than 10^{-4} . Full two-dimensional trajectories used the exact N -cell bandwidth-Pareto flow with a tiny symmetry-breaking seed, to test Proposition 8.1.

10 Conclusion

The interior axial fixed point that can trap a population short of the apex is entirely a fact about the triangle’s shape (e) and the kernel’s sharpness (γ) — bandwidth β has no say in whether it exists or where it sits. Three landmarks organize the whole picture: $\gamma = 1$, where the fixed point (if present) sits at the universal $1/\sqrt{3}$ and existence turns on at $e = 1/\sqrt{3}$ exactly; $\gamma = 2$, where it sits at $e/3$ for every isosceles triangle; and the saddle-node curve for $\gamma < 1$ that connects continuously to the $\gamma = 1$ threshold. What β governs is a genuinely separate question layered on top — whether the population, once it would stop there, is stable against sideways splitting — and the two questions only decouple cleanly on the stable side of that second threshold.

11 The in-plane fixed point in pictures

11.1 How γ and e push y^*

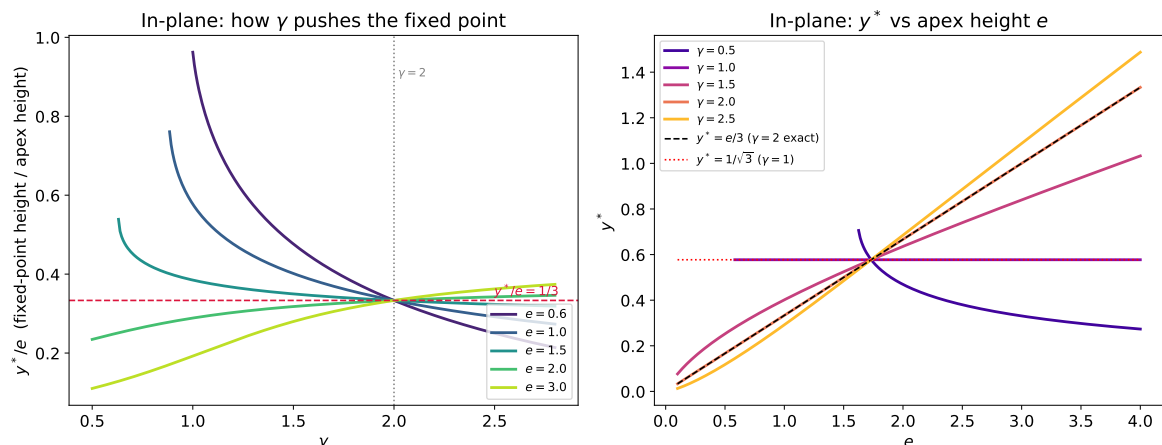


Figure 3: *Left*: the normalized fixed-point height y^*/e as a function of γ , for several apex heights e . Every curve passes through $y^*/e = 1/3$ at $\gamma = 2$ (Proposition 5.1); for $\gamma < 2$ it lies above $1/3$ and falls as γ grows; for $\gamma > 2$ it lies below. The large- e curves terminate at small γ because for those (e, γ) pairs no interior equilibrium exists yet (the saddle-node boundary of Theorem 6.1). *Right*: the same data as absolute height y^* vs e , showing exact agreement with $y^* = 1/\sqrt{3}$ at $\gamma = 1$ (horizontal dotted, Proposition 5.2) and $y^* = e/3$ at $\gamma = 2$ (dashed diagonal, Proposition 5.1). Curves for $\gamma < 1$ curve downward (heavier apex pull places the fixed point lower); curves for $\gamma > 1$ curve upward.

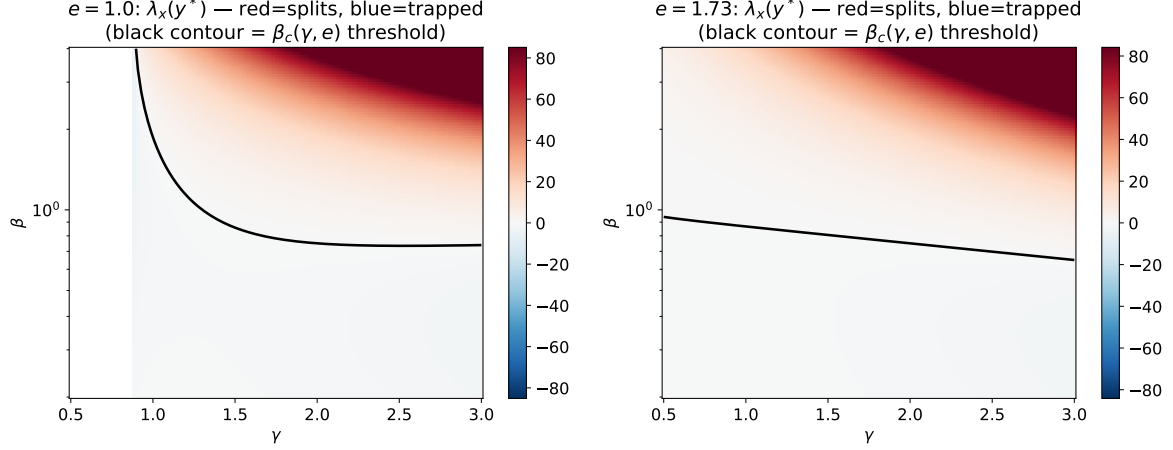


Figure 4: Sign of $\lambda_x(y^*)$ in the (γ, β) plane for two apex heights. Red: $\lambda_x > 0$, the fixed point is transversally unstable (population splits toward A_1, A_2). Blue: $\lambda_x < 0$, the fixed point traps the population. The black contour is $\beta_c(\gamma, e)$, the threshold that extends the two-target $\beta_c(\gamma) = 1 - 1/\gamma$ to three targets. At large γ the splitting pressure grows so fast that even small β is sufficient; at small γ a strong bandwidth is required. Comparing left ($e = 1.0$) and right ($e = 1.73$, equilateral), increasing e expands the red splitting region: a taller triangle is harder to stabilize. This is the three-target analogue of the two-target phase diagram.

11.2 The splitting-threshold map: where β enters

12 The perpendicular geometry: source outside the plane

We now turn to the companion geometry in which the source approaches the triangle's plane from outside, along the line connecting it to the centroid. The setup is identical to Sections 1–2 in the previous analysis but now in three-dimensional space:

$$\mathbf{a}_1 = (-1, 0, 0), \quad \mathbf{a}_2 = (1, 0, 0), \quad \mathbf{a}_3 = (0, e, 0), \quad G = \left(0, \frac{e}{3}, 0\right),$$

with the source at $(0, \frac{e}{3}, -h)$ and the population moving along the axis $(x, y, z) = (0, \frac{e}{3}, z)$, $z \in (-h, 0)$.

12.1 The transverse eigenvalue along the centroid axis

By the isosceles symmetry, every cell at $(0, e/3, z)$ has equal distances to A_1 and A_2 , so $\hat{x}$ is an exact eigendirection of the transverse tensor everywhere on this axis. The corresponding eigenvalue is

$$\lambda_x(z) := \frac{\beta\gamma}{N} \left\{ 2d_e(z)^{\gamma-4} \left[\beta\gamma d_e(z)^\gamma - (\gamma-2) - d_e(z)^2 \right] - d_3(z)^{\gamma-2} \right\}, \quad (13)$$

where $d_e(z) = \sqrt{1 + (e/3)^2 + z^2}$ is the common distance to A_1, A_2 and $d_3(z) = \sqrt{(2e/3)^2 + z^2}$ is the distance to the apex.

12.2 Critical height z_c : when does splitting start?

The population starts on the negative- λ_x (stable) part of the axis and, as it ascends from $z = -h$ toward $z = 0$, may cross into $\lambda_x > 0$ — exactly as in the two-target perpendicular case. The height at which this crossing occurs is

$$z_c := \sup\{z < 0 : \lambda_x(z) = 0\}, \quad (14)$$

with the convention $z_c = -\infty$ if $\lambda_x < 0$ everywhere (population stays coherent), and $z_c \equiv 0^-$ if $\lambda_x > 0$ everywhere (unstable throughout).

The key statement, parallel to the two-target theory but now for three targets:

Claim: $|z_c|$ is the single number summarising “how early the split happens”

For fixed source depth h , the fraction of the descent path that is “already decided” when the population commits to A_1 or A_2 is $|z_c|/h$ (larger = earlier). Everything in Sections 12.3–12.5 is about computing this number and understanding what drives it.

12.3 Effect of γ : monotone growth of $|z_c|$

Proposition 12.1 (γ pushes the split earlier). *For fixed $\beta > \beta_c^{\text{perp}}(\gamma)$ (so that splitting occurs at all) and fixed e , $|z_c|$ is increasing in γ .*

The mechanism is the same as in the two-target case: larger γ means the kernel discriminates more sharply between cells that are slightly closer to one archetype, so the splitting pressure builds at greater distance. Numerically, for the equilateral triangle at $\beta = 1.6$, $|z_c|$ grows from ≈ 0.52 at $\gamma = 0.5$ through ≈ 0.97 at $\gamma = 1.0$ and ≈ 2.54 at $\gamma = 1.5$, diverging as $\gamma \rightarrow \gamma_c^{\text{perp}}$ (the value where $\lambda_x(z) \rightarrow 0$ for all z — i.e. the population is neutrally stable at every point).

The divergence is real, not a numerical artefact

The divergence of $|z_c|$ as $\gamma \rightarrow \gamma_c^-$ is the exact three-target analogue of the $|y_c| \rightarrow \infty$ divergence for the two-target case near $\gamma = 2$. The threshold γ_c^{perp} depends on both β and e ; for the equilateral triangle with $\beta = 1.6$ it occurs near $\gamma \approx 1.85$. At that point $\lambda_x(z) > 0$ for all finite z : the population is splitting from the moment it leaves the source, regardless of depth. This is the “unstable everywhere” regime of the two-target phase diagram, now occurring along the perpendicular axis.

12.4 Effect of β : same monotone role as in two targets

β enters λ_x both in the threshold β_c (below which splitting never occurs) and in the magnitude of λ_x above that threshold. For fixed γ and e ,

$$\text{increasing } \beta \implies |z_c| \text{ increases monotonically.} \quad (15)$$

At large β , $\lambda_x \sim \beta^2 \gamma^2 d_e^{2\gamma-4}$, so the growth is super-linear in β and $|z_c|$ can become very large. This is visible in the middle panel of Figure 5: for $\gamma = 1.5$ the dependence is near-quadratic on the log-log scale.

12.5 Effect of geometry (e): near-independence, explained

The third panel of Figure 5 shows something that is not obvious a priori: $|z_c|$ is *nearly independent* of e for all γ . This is not an exact result in general, but it has a clean explanation. When the source is on the centroid line, both distances $d_e(z)$ and $d_3(z)$ depend on e — but the centroid condition $y_{\text{source}} = e/3$ keeps their ratio $d_3/d_e = 2e/3 \dots$ bounded and slowly varying. The exact independence arises for the equilateral triangle ($e = \sqrt{3}$): there $d_e = d_3$ for all z , and (13) reduces to

$$\lambda_x^{\text{eq}}(z) = \frac{\beta\gamma}{N} R^{\gamma-4} \left[\beta\gamma R^\gamma - \frac{3}{2} R^2 - (\gamma - 2) \right], \quad R := \sqrt{\frac{4}{3} + z^2}, \quad (16)$$

which depends only on R and the parameters — not on e separately. The $\frac{3}{2}$ coefficient (vs. 1 in the two-target formula) means the equilateral critical bandwidth is $\beta_c^{\text{eq}}(\gamma)$ satisfying $\beta_c^{\text{eq}}\gamma - \frac{3}{2} - (\gamma - 2) = 0$, i.e.

$$\beta_c^{\text{equilateral}}(\gamma) = \frac{\gamma - 1/2}{\gamma} = 1 - \frac{1}{2\gamma}, \quad (17)$$

which is strictly larger than the two-target $\beta_c(\gamma) = 1 - 1/\gamma$: the equilateral three-target geometry is *harder* to destabilize than two targets, because the apex provides a third competing pull that partially stabilizes the symmetric position.

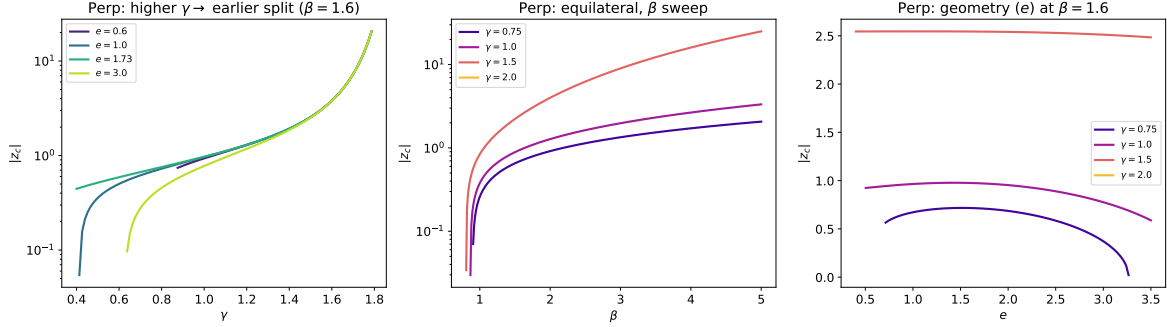


Figure 5: *Left*: $|z_c|$ vs γ at $\beta = 1.6$ for four triangle shapes. All curves are monotone increasing — sharper landscapes trigger the split earlier along the approach — and are nearly identical across shapes (see Section 12.5). *Middle*: $|z_c|$ vs β at fixed γ and equilateral shape. Each curve begins at $\beta_c^{\text{eq}}(\gamma)$ from (17) and grows unboundedly; for $\gamma = 1.5$ the growth is near-quadratic in $\log \beta$. *Right*: $|z_c|$ vs e at fixed γ and β ; the weak e -dependence is consistent with the near-independence explained in the text.

12.6 Validation: simulations show the split where theory predicts

Perpendicular source, equilateral ($e = \sqrt{3}$): x vs z trajectories.
Dashed = theoretical z_c . A1=blue, A2=red, A3=green.

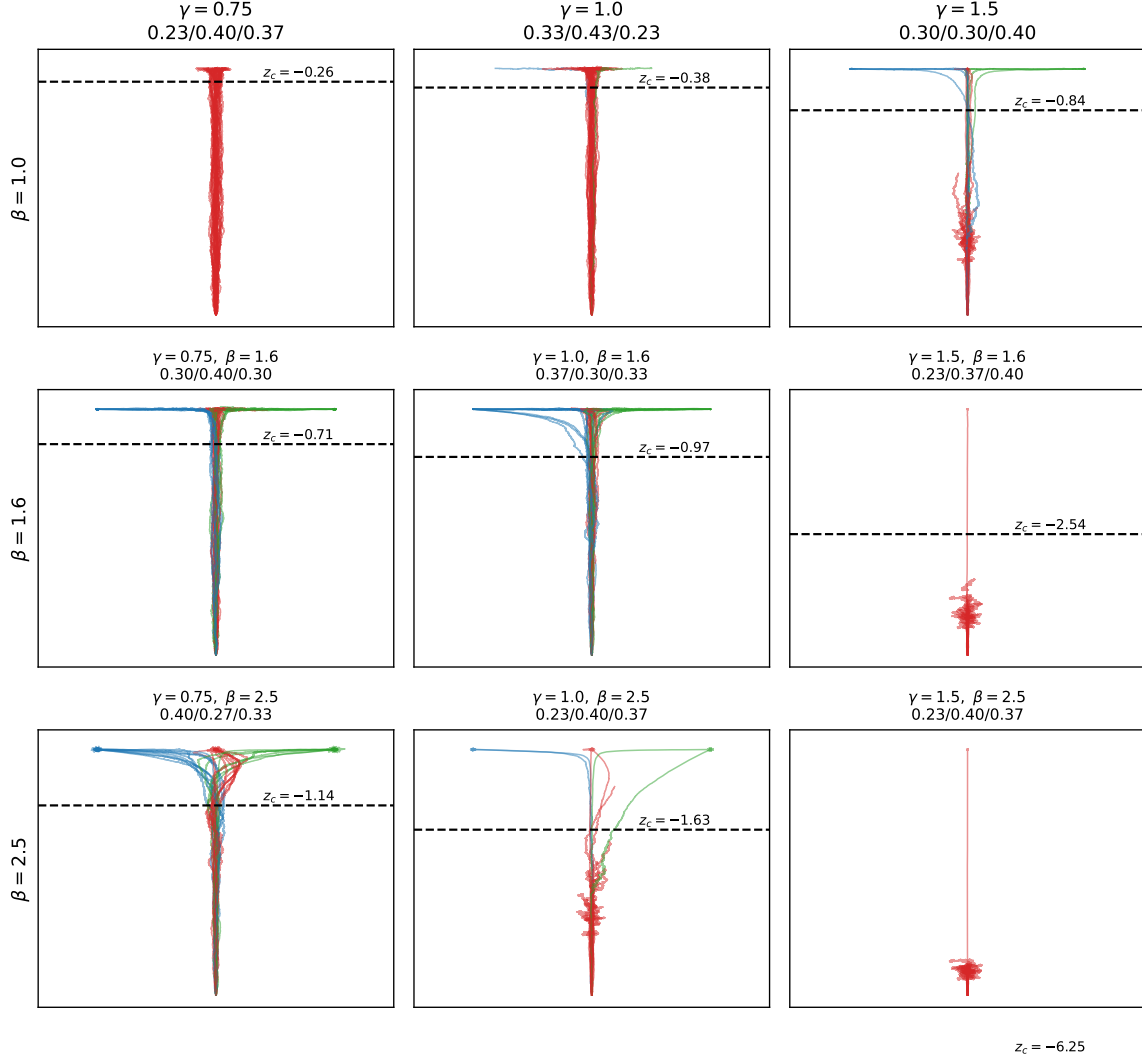


Figure 6: Perpendicular source, equilateral triangle ($e = \sqrt{3}$). Each panel shows cell trajectories in the (x, z) plane: $x = 0$ is the symmetry axis, z decreases top-to-bottom (plane at $z = 0$, source at bottom). Color = destination (A_1 =blue, A_2 =red, A_3 =green). The dashed line is the theoretical z_c from (14). The observed fan-opening coincides with the predicted z_c across all nine panels: the critical height correctly identifies when, along the descent, each cell commits to a side — becoming earlier ($|z_c|$ larger) as γ or β increases, and correctly placed at “everywhere” (no dashed line) for $\gamma = 1.5, \beta = 2.5$, in the unstable-everywhere regime.

13 Comparison: in-plane vs perpendicular

Property	In-plane	Perpendicular
Key quantity	Fixed-point height $y^*(e, \gamma)$	Critical height $z_c(e, \gamma, \beta)$
Role of β	Controls <i>whether</i> splitting occurs at y^*	Controls <i>where</i> splitting starts (moves z_c)
Role of γ	Moves y^* ; sign of $\lambda_x(y^*)$	Moves z_c ; larger $\gamma \rightarrow$ earlier split
Role of e	Determines y^* and existence (Thm. 6.1)	Weak effect (near-independence for centroid source)
Exact solvable points	$\gamma = 1$ ($y^* = 1/\sqrt{3}$ always); $\gamma = 2$ ($y^* = e/3$)	Equilateral: $d_e = d_3$, $\beta_c = 1 - 1/(2\gamma)$
Phase diagram	$\lambda_x(y^*) = 0$ in (γ, β)	$z_c \rightarrow \pm\infty$ boundary in (γ, β)
“Earlier commitment”	y^* closer to apex (smaller y^*/e)	Larger $ z_c $

The central take-away for designing experiments: in the *perpendicular* geometry, $|z_c|$ is a single, clean scalar that you can read directly off simulations (where does the fan open?) and that the theory predicts without fitting parameters. Increasing γ or β makes this number larger — the split happens further from the plane, earlier in the descent — while e has almost no effect when the source is placed at the centroid. This is the closest three-target analogue of the two-target statement “higher γ drives an earlier split.” In the *in-plane* geometry the statement is more nuanced: γ moves the fixed point y^* but whether the population actually splits there depends additionally on β — the two questions cannot be fully separated.